

Should I State or Should I Show? Aligning AI with Human Preferences*

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Abstract

We study how effectively an AI agent aligns with a human principal’s choices under risk when given *stated* versus *revealed* preference information. We conduct an online experiment in which subjects state their preferences through written instructions (“prompts”) and reveal them through a series of binary lottery choice problems (“data”). We find that on average, an AI agent given data predicts subjects’ choices more accurately than an AI agent given prompts. We attribute this gap to subjects’ difficulty in translating their own preferences into written instructions. Subjects also misperceive the relative performance of the two agents and delegate to the less accurate agent too often. Moreover, providing AI agents with more information does not necessarily improve their performance, since different AI models resolve conflicts between stated and revealed preferences differently. Overall, our results highlight revealed preferences as a powerful mechanism for communicating human preferences to AI agents, but their success depends on careful implementation.

1 Introduction

The emergence of agentic artificial intelligence (AI) has prompted widespread discussion of a future in which AI agents act autonomously on behalf of humans, with little or no direct oversight. Although views may differ on the exact form this AI-assisted

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future will take, there is a growing scholarly consensus that AI will fundamentally transform economic interactions and productivity.¹ At the same time, the proliferation of AI agents introduces a new form of principal-agent problem in which misalignment may arise not because AI agents have preferences of their own, but because human principals cannot fully articulate their preferences.²

As a simple example, consider a traveler, the principal, who tasks an AI agent with “buying the cheapest flight from Melbourne, Australia to Washington, D.C.” The agent may select an itinerary with an extremely long layover—an outcome the principal might implicitly wish to avoid but did not explicitly rule out in their instructions. Furthermore, the decisions made by the AI agent may be irreversible and costly, e.g., the purchase of a cheaper but nonrefundable flight ticket, as they take on a more autonomous role in decision-making (Immorlica et al., 2024). More generally, individuals may be unable to fully specify their preferences and such incompleteness in preference specification is likely to persist. Thus, a first-order challenge is to design mechanisms that mitigate the resulting misalignment.

To address this challenge, in this paper, we adopt a *revealed preference* approach that uses the principal’s past choices as an alternative source of information. Observed choice data provides a direct channel through which human principals communicate their preferences to an AI agent. We compare this approach with one based on *stated preferences*, elicited through incentivized instructions (“prompts”) that human principals write to guide the AI agent’s decisions on their behalf. The question we ask is: which approach better enables an AI agent to align with and implement the human principal’s preferences in the domain of choice under risk?

On the one hand, if human principals can precisely articulate their preferences, then an AI agent given those preferences may be able to implement them across a wide range of decision problems. On the other hand, human principals may be unable to articulate their preferences as precisely as they are revealed through their choices, making it harder for the AI agent to act on stated preferences alone. Thus, a priori, it is unclear which approach yields a better alignment.

To study this question, we conduct an incentivized online experiment that elicits

¹For some recent work, see, e.g., Brynjolfsson et al. (2025); Acemoglu (2025); Gans (2026).

²This phenomenon has recently been referred to as *specification hazard*; see Imas et al. (2025); Shahidi et al. (2025). In contrast, misalignment arises in the classical economic setting because the principal and the agent have conflicting incentives. This is the well-known moral hazard problem, where the agent’s action is imperfectly observable, and the principal’s problem is to design a contract that aligns the agent’s incentives with their objectives (Holmström, 1979).

both revealed and stated preferences in a choice-under-risk setting. The experiment consists of two main parts. In Part I, subjects answer a series of binary choice problems over lotteries with varying levels of difficulty. They are then asked to write a prompt stating their preferences to an AI agent that will act on their behalf in a new series of similar lottery choice problems. In Part II, subjects then make choices in these new problems. For each subject, we instantiate two AI agents using Anthropic’s Claude Opus 4.5—a frontier large language model at the time of writing. The first agent, **Data-AI**, is given the subject’s choices in Part I; the second agent, **Prompt-AI**, is given the subject’s written prompt. We evaluate the performance of these agents by comparing their predictions with subjects’ actual choices in Part II, using *match rate*—out-of-sample prediction accuracy—as our metric.

Overall, we find that on average, AI agents perform significantly better when given revealed preferences rather than stated preferences. At the same time, there is marked heterogeneity in the performance gap across subjects: subjects who more frequently exhibit Allais-type behavioral biases (Allais, 1953) are harder to predict for both types of AI agents. This pattern is especially pronounced for Prompt-AI, which performs as well as Data-AI for subjects who exhibit no such biases but performs six percentage points worse for subjects who exhibit these biases most frequently. Thus, the subjects whose choices are least consistent with the canonical expected utility framework are precisely those who benefit most from providing revealed-preference information rather than stated-preference information.

We next investigate why Data-AI performs better. We provide three pieces of evidence suggesting that the performance gap is driven by subjects’ difficulty in articulating their preferences in written prompts. First, we find that subjects’ written instructions only partially predict the choices they made *before* writing the prompt. Second, we instead use AI to generate prompts using the same instructions and information provided to subjects and then instantiate another AI agent using these prompts. The resulting AI agents are substantially more predictive, both within-sample and out-of-sample, with performance on par with Data-AI. Third, as subjects’ written prompts become semantically similar to those generated by AI, the performance gap between Data-AI and Prompt-AI essentially closes.

Building on the observed performance gap between AI agents, we then study whether human principals recognize this gap when choosing between the two agentic regimes and the welfare consequences of their choices. In the experiment, after sub-

jects make their choices and write their prompt, we also ask them to choose exactly one of the two agents (Data-AI or Prompt-AI) to delegate decisions to. The chosen agent then makes choices on the subject’s behalf in Part II of the experiment. We find that 61% of subjects delegate to Data-AI. Their delegation choices also largely reflect their perceived AI performance: overall, 82% of subjects choose the agentic regime they believe to be weakly better. However, subjects substantially overestimate the true absolute difference in match rates between these agents, and are often mistaken about which agent performs better; as a result, a substantial share (36%) of subjects fail to choose the better-performing agent.

Finally, we investigate how an AI agent behaves when given both stated and revealed preference information; we refer to this agent as **Both-AI**.³ Intuitively, giving more information to the AI agent should at least weakly improve its performance. However, we find instead that Both-AI performs significantly *worse* than Data-AI and only marginally better than Prompt-AI. This performance reduction is driven by the 26% of subject-question pairs in which Prompt-AI and Data-AI make *conflicting* predictions; within this subset, Both-AI follows Prompt-AI’s prediction 66% of the time, despite Prompt-AI being less accurate on average.

We note that the above pattern is model-specific. A GPT-based Both-AI agent performs as well as Data-AI, precisely because it follows the better-performing agent, Data-AI, more frequently on conflicting questions. This difference is consistent with differences in how these two large language models are trained: Anthropic emphasizes constitutional-style alignment (Bai et al., 2022), which explicitly prioritizes the stated goals of a human user whereas OpenAI uses a more standard reinforcement learning from human feedback approach (Ouyang et al., 2022). Nonetheless, when we use GPT-5.4 as the underlying model, the rest of our main results remain both qualitatively and quantitatively similar across the different AI models.

Taken together, our findings suggest that revealed preference can be an effective approach for human principals to communicate their preferences to an AI agent. However, this approach may also require careful implementation: humans may fail to opt into using this information on their own, and overloading AI agents with information may reduce their predictive performance.

³Specifically, for each subject, we instantiate a third AI agent endowed with both the subject’s choices and written prompt in the first part of the experiment.

2 Related Literature

Our paper contributes to a burgeoning literature on AI agents as economic actors.⁴ A central concern in this literature is the potential misalignment between an AI agent’s objective and that of its human principal, for instance, when the two have different preferences (Gabriel, 2020; Ji et al., 2023; Potter et al., 2026; Suleymanov, 2026). Beyond inherent preference differences, misalignment can also arise from technical limitations of AI agents, such as inaccurate inference or hallucination (Huang et al., 2025b), or from the human principal’s inability to fully articulate their preferences to the AI agent. We focus on this latter channel, which we view as the primary source of misalignment in our setting.⁵

As is well known, prompt writing is hard, and there is no reliably effective universal strategy (Zamfirescu-Pereira et al., 2023; Meincke et al., 2025a,b,c). Recent evidence from Imas et al. (2025) also shows substantial heterogeneity in human principals’ ability to state their preferences, with these differences carrying through to economic outcomes when decisions are implemented by AI agents. This source of misalignment may become even more severe in high-dimensional decision problems, where human principals could face greater difficulty articulating their preferences precisely (Shahidi et al., 2025; Liang, 2026). Our work shows that revealed preference information, as an alternative channel for communicating human principals’ preferences, can help mitigate this form of misalignment.

Our paper is related to the line of work that uses LLMs to extract preferences from natural language for economic applications. For example, Rusak et al. (2025) show that LLMs can convert free-text taste descriptions over job roles into cardinal utilities that capture human subjects’ preferences, thereby improving allocation mechanisms in a labor-market matching experiment. In an auction setting, Huang et al. (2025a) show that LLM-powered proxies can help determine which revealed-preference information to query from humans. Furthermore, Li et al. (2023) show that such stated preference elicitation can be improved with a framework that allows active feedback

⁴See recent surveys, e.g., Immorlica et al. (2024) and Hadfield and Koh (2025).

⁵A growing literature supports this view, showing that LLMs exhibit consistent decision-making in choice under risk contexts (Chen et al., 2023). For example, Kim et al. (2024) show that GPT’s recommendations are consistent with expected utility maximization and can be aligned with subjects’ risk aversion when provided with simulated choice data. Moreover, broader AI alignment concerns are unlikely to affect our results: our tasks involve standard choices under risk, with no safety (Amodei et al., 2016; Hendrycks et al., 2023) or constitutional-AI constraints (Bai et al., 2022) that prevent the AI agent from implementing the risk preferences stated by human principals.

and interaction between the language models and humans. Rather than studying how to improve preference elicitation through natural language, we compare stated and revealed preferences as alternative inputs for enabling AI agents to implement human principals’ preferences in a controlled and incentivized environment.⁶

More broadly, our work is also related to the literature on human-AI interaction.⁷ Within this literature, we are most closely related to studies examining human beliefs about AI capabilities and their effects on delegation decisions. For instance, [Vafa et al. \(2024\)](#) show that overestimation of AI performance can lead to worse outcomes because humans delegate too much. Similarly, [He et al. \(2025\)](#) find that humans believe AI agents’ behavior is far closer to their own than it actually is, while [Dreyfuss and Raux \(2025\)](#) find that these beliefs update more sharply in tasks where AI performance does not conform to human expectations (e.g., by performing poorly in a “human-easy task”). We contribute to this literature by examining human subjects’ beliefs about the relative performance of AI agents constructed from different information sources, and by showing that the resulting misperceptions also matter for effective human-AI interaction.

3 Experimental Design

Structure of the Experiment. The experiment consists of three parts: Part I and II are the main components of the experiment, in which subjects make choices under risk, write instructions for an AI agent, and decide what type of information they would like to provide to an AI agent that will later act on their behalf.

In Part I of the experiment, subjects complete three tasks. They are informed that their decisions in each task will be used to instantiate an AI agent. This agent will then make decisions on their behalf in a new but structurally similar set of lottery problems that may be used to determine their bonus payment. In the first task, subjects choose between 13 pairs of binary lotteries. These lottery problems are classified into three different categories—“easy”, “hard”, and “behavioral” (more detail below)—and all are presented in random order. Subjects also have access to

⁶Other papers have also collected stated and revealed preferences in an unincentivized manner, such as [Fedyk et al. \(2024\)](#) and [Lai et al. \(2026\)](#). However, they elicit stated preferences only in the more limited form of Likert scales rather than richer free-text responses, and they do not examine LLM responses to the two information sources separately. They also study different settings: the former augments LLMs with demographic information, while the latter focuses on writing assistance.

⁷For a recent survey, see [Jackson et al. \(2025\)](#).

five summary statistics about each lottery to assist their decision-making.⁸ We refer to the resulting AI agent endowed with this choice information as **Data-AI**.

In the second task, after completing their choices, subjects write a free-form prompt describing their preferences for choosing between lotteries, which will then be given to an AI agent. Subjects are encouraged and incentivized to provide clear, flexible guidance that can be used across different situations, but do not explicitly provide prompt examples to prevent anchoring (Furnham and Boo, 2011). While writing their prompt, subjects have access to their previous choices, including the summary statistics of the lotteries. Finally, to limit the opportunity cost of time spent on the prompt, a minimum of 60 seconds is enforced on the prompt-writing screen before subjects can progress (Spiliopoulos and Ortmann, 2018). We refer to the resulting AI agent endowed with the prompt as **Prompt-AI**.

In the third task, subjects decide which type of information to provide to the AI agent instantiated from this task. They must select exactly one of two options: their *written instructions* from task 2 or their *past choice data* from task 1. We refer to this as their “delegation decision.”

In Part II of the experiment, subjects now face the new set of 13 binary lottery pairs that are designed to be structurally similar to those in Part I (see more details on the lottery specifics below). Subjects’ choices in these problems then serve as the ground truth for evaluating AI agent predictions. However, subjects are not informed, at the time they make their Part II decisions, that these choices will be used to benchmark AI accuracy. This avoids subjects potentially choosing in accordance with their written prompts or past choices, which may not always correspond to their true preferences. In both Part I and Part II, a brief brain break is implemented after every six binary lottery questions to reduce fatigue. During these breaks, subjects are asked to locate a camouflaged animal in two distinct images.

Part III of the experiment consists of supplemental measures. We elicit subjects’ beliefs about the prediction accuracy of Prompt-AI and Data-AI agents. Subjects are incentivized to correctly guess how many questions each type of AI agent predicts correctly. We also ask subjects why they made their delegation choice. Subjects also complete a series of control tasks, containing a short IQ test (ICAR, Condon and Revelle, 2014), risk attitude elicitation using two investment tasks (Gneezy and Pot-

⁸These statistics are the expected value (displayed as “Average Payment”), the variance (displayed as “Payment Variability”), the minimum payment, the maximum payment, and the probability of receiving the maximum payment. See Arrieta and Nielsen (2025) for a similar implementation.

ters, 1997), and measures of overconfidence. At the end of the experiment, subjects answer a brief, unincentivized survey covering self-reported comfort with AI tools, literacy, impatience, frequency of AI use, education level, and management experience, as well as a personality survey (TIPI; Gosling et al., 2003). We additionally collect demographic variables such as age and gender.⁹

Incentives. The payment of subjects consists of several parts. First, subjects receive a \$5 completion payment. Second, to incentivize subjects to write their prompt carefully and to select the information they prefer to send to the AI to act on their behalf, one of four sets of 13 binary lottery choices is randomly selected for the bonus payment. Three of these sets consist of AI predictions for the Part II lottery problems, generated using different information sources from Part I: one from Data-AI based on the subject’s past choices, one from Prompt-AI based on the subject’s written prompt, and one from either based on the subject’s delegation decision. The fourth set consists of the subject’s own 13 choices from Part II.¹⁰

From the selected set, *one* decision is drawn at random, with equal probability across the 13 problems, and the corresponding lottery is implemented for payment. This procedure ensures that subjects have incentives to make genuine choices, write informative prompts, and delegate to the AI agent so that it can best act on their behalf. Finally, subjects are paid for almost all decisions made in Part III, except the brief, unincentivized summary survey at the end. The experiment lasted on average 27 minutes. The average total earnings per subject were just over \$12, implying an hourly rate of approximately \$27 per hour.

Implementation. The experiment was programmed in oTree (Chen et al., 2016) and administered via Prolific across three independent sessions (February, March, and April 2026), with recruitment restricted to U.S.-based adults. A total of 296 subjects passed all comprehension quizzes and completed the experiment. For our analysis, we excluded six subjects whose written prompts, upon close inspection, were completely uninformative about their preferences.¹¹ In addition, to ensure that prompts were

⁹See Appendix D for more details on these supplemental measures.

¹⁰Note that subjects are incentivized based on the choices the AI agent actually makes on their behalf, rather than on whether those choices match with the subjects’ own Part II choices. This avoids concerns that subjects might adopt a more easily articulated decision rule, rather than choosing according to their true preferences.

¹¹As an example, two of them submitted empty prompts, and one wrote “select A”.

written by human subjects, we also excluded 23 subjects for whom we detected that content had been pasted into their written prompts. This left us with 267 subjects in total.

All predictions made by the AI agent were generated using Claude Opus 4.5 (`claude-opus-4-5-20251101`), with extended thinking enabled. Each agent (Prompt-AI or Data-AI) is prompted to reason through each lottery pair and submit its choices in a standardized format. The full system prompts used for each agent are reproduced in Appendix B. The study was approved by the Monash Human Research Ethics Committee (ID: 50731), and was pre-registered on AsPredicted (# 268985). Instructions and screenshots of the interface are presented in the Online Appendix.

The Lottery Choices. Human subjects face two sets of 13 lottery pairs, one in Part I and one in Part II. Within each set, the 13 pairs span three conceptually distinct types of decisions: “easy”, “hard”, and “behavioral”, which are shown in Table A.1 in Appendix A. Easy questions involve either first-order stochastic dominance or large differences in expected value between the two lotteries. Hard questions, in contrast, involve lotteries with smaller expected value differences, more outcomes on average, no obvious heuristic solution, or require reasoning based on more subtle dominance concepts such as second-order stochastic dominance. Finally, behavioral questions are designed to test for (reverse) common ratio and (reverse) common consequence effects—two canonical environments in which violations of expected utility theory are frequently observed empirically (Blavatskyy et al., 2022, 2023).

We use two easy questions and five hard questions in both parts. In Part I, we source one easy question and four hard questions from Agranov and Ortoleva (2017).¹² The remaining easy and hard questions compare first- and second-order dominated lotteries, respectively. In Part II, we increase the stakes tenfold for one easy question and one hard question; for the remaining questions we introduce slight perturbations while keeping expected value differences largely unchanged.

In each part, we also generate two sets of three behavioral questions using the framework of McGranaghan et al. (2026). Specifically, for fixed prizes $H > M > 0$, consider the following three binary choice questions that are parameterized by a vector

¹²In Agranov and Ortoleva (2017), lottery outcomes are reported in tokens rather than dollar amounts. We convert those token values into dollars and present all outcomes to subjects directly in monetary terms.

(p, r) where $p, r \in (0, 1)$:¹³

- (i) AB choice: choose lottery $A = (M, 1)$ or lottery $B = (H, p)$.
- (ii) AB' choice: choose lottery $A = (M, 1)$ or lottery $B' = (H, pr; M, 1 - r)$
- (iii) CD choice: choose lottery $C = (M, r)$ or lottery $D = (H, pr)$.

Under expected utility theory, multiplying the probabilities of non-zero outcomes by a common factor r , or replacing a shared consequence of a $1 - r$ probability of $\$M$ with a $1 - r$ probability of $\$0$, should not change preferences. However, these factors are routinely observed to affect choices. The common ratio effect is identified by a preference reversal between the AB choice and CD choice: individuals prefer A to B in the AB task but then prefer D to C in the CD task. The common consequence effect is identified by an analogous preference reversal between AB' and CD . The reverse of these effects, intuitively, reverse choice in each question.

For the behavioral problems in Part I, we choose $(p, r) = (8/10, 1/4)$ and $(p, r) = (1/2, 1/2)$. For the behavioral problems in Part II, we choose $(p, r) = (10/11, 11/100)$ and $(p, r) = (3/10, 1/2)$. We additionally eliminate certainty for option A in Part II, instead using $A = (M, 0.95)$ to eliminate the certainty effect. These parameterizations are chosen to generate a variety of common ratio, reverse common ratio, common consequence, and reverse common consequence patterns (Allais, 1953; Kahneman and Tversky, 1979; McGranaghan et al., 2026).¹⁴

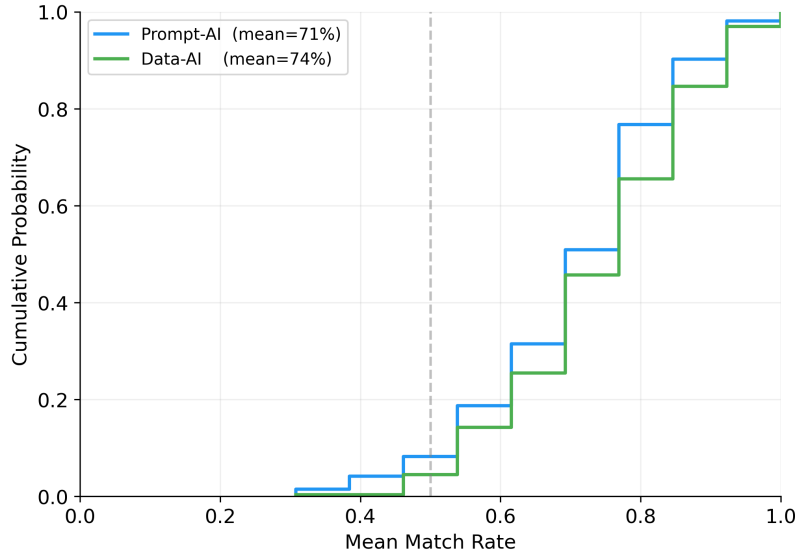
4 Results

Our primary outcome of interest is the AI agent’s *match rate*, which is a subject-level, out-of-sample measure defined as the fraction of Part II decisions for which the AI agent’s prediction coincides with the subject’s actual choice. We compare the match rates of Prompt-AI and Data-AI and examine the sources of discrepancies between

¹³Here, lotteries are represented as $(x_1, p_1; \dots, x_n, p_n)$ where each outcome x_i occurs with probability $p_i \geq 0$ and $\sum_{i=1}^n p_i = 1$. To simplify notation, we omit the null outcome of 0 and use (x, p) to denote the prospect $(x, p; 0, 1 - p)$.

¹⁴In Part I, the former set of parameters is expected to generate a common ratio effect but no common consequence effect, while the latter is expected to generate strong reverse common ratio and strong reverse common consequence effects. Likewise, in Part II, the former is expected to generate strong common ratio and common consequence effects, while the latter is expected to generate only a reverse common consequence effect.

FIGURE 1: Comparison of Prompt-AI and Data-AI match rates.



Note: The empirical CDF of per-subject match rates are shown for Prompt-AI (blue) and Data-AI (green). The dashed line at 50% is the expected match rate of random guessing. Error bars show 95% confidence intervals with standard errors clustered at the subject level. Stars denote paired t -tests comparing Data-AI to Prompt-AI within each category. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

them. We then study subjects’ delegation choices. Specifically, when subjects are given a choice between the two AI agents, we examine whether they delegate to the agent with the higher match rate and whether their delegation choices align with their perceptions. Finally, we study how AI agents handle conflicting predictions from stated and revealed preferences.

4.1 Revealed versus Stated Preferences

We find that Data-AI systematically outperforms Prompt-AI across subjects and across question categories. At the aggregate level, Data-AI achieves an average match rate of 74%, which is three percentage points higher than the 71% achieved by Prompt-AI (paired t -test: $t = 3.69$, $p < 0.001$). This advantage is also present across all three question categories discussed in Section 3.¹⁵ In Figure 1, we present the

¹⁵The average match rates for Data-AI are 99%, 70%, and 69% for easy, behavioral, and hard questions, respectively. The corresponding average match rates for Prompt-AI are 95%, 66%, and 67%, respectively. Paired t -tests reject the null hypothesis of equal match rates for easy and behavioral questions, and marginally so for hard questions ($p < 0.001$, $p = 0.014$, and $p = 0.097$, respectively).

empirical CDFs of match rates for Prompt-AI and Data-AI and make two observations. First, regardless of the AI agent, subject-level match rates are nearly all to the right of the dashed line at 50%, indicating that both AI agents effectively incorporate the preference information they are given. Second, while there is heterogeneity in match rate at the subject level, the distribution of Data-AI match rates first-order stochastically dominates that of Prompt-AI.

To benchmark the performance of these AI agents in our setting, we empirically estimate an expected utility (EU) model with CRRA risk preferences using subjects' choices in Part I. We then use the estimated risk parameters to predict subjects' choices in Part II (see Appendix C for details). We find that on average, the EU model achieves a match rate of 75%, which is statistically indistinguishable from that of Data-AI (paired t -test: $t = 1.05$, $p = 0.295$). At the subject-question level, Data-AI also aligns more closely with the EU model than Prompt-AI does, coinciding with the EU prediction 80% of the time, compared to 73% for Prompt-AI. Thus, the predictive performance of Data-AI is comparable to that of a standard structural model of risk preferences.

To examine who benefits more from providing the AI agent with revealed-preference information, we classify subjects based on their answers to the two sets of three behavioral questions. Specifically, subjects are classified based on the number of times they exhibit the (reverse) common ratio or (reverse) common consequence effects. Note that each set of behavioral questions contains the possibility of one (reverse) common ratio effect and one (reverse) common consequence effect. Since we have two sets of behavioral questions, in total there are four possible observable effects. Overall, 25% of subjects exhibited none of these behavioral patterns, 52% of subjects exhibited one or two, and 23% of subjects exhibited three or four.

Table 1 shows the relationship between match rate and behavioral effects. The dependent variable is an AI agent's mean match rate in Part II. Column (1) regresses match rate on an indicator for whether the agent is Data-AI or not. Column (2) adds the number of behavioral patterns exhibited by the subject, and column (3) further includes its interaction with the Data-AI indicator. Column (4) adds controls for demographics and survey responses that we elicited in Part III of the experiment.¹⁶

As before, we observe that Data-AI outperforms Prompt-AI by three percentage points on average. Subjects who exhibit more behavioral patterns also have lower

¹⁶See Table D.1 in Appendix D for the full regression table.

TABLE 1: OLS: Match Rate

	Match rate			
	(1)	(2)	(3)	(4)
Data-AI (vs Prompt-AI)	0.033*** (0.009)	0.033*** (0.009)	0.013 (0.014)	0.013 (0.014)
Behavioral Effects		-0.032*** (0.006)	-0.038*** (0.007)	-0.040*** (0.008)
Data-AI $\times$ Behavioral Effects			0.012* (0.007)	0.012* (0.007)
Constant	0.707 (0.009)	0.758 (0.013)	0.767 (0.014)	0.770 (0.014)
Controls	No	No	No	Yes
Observations	532	532	532	532
Subjects	266	266	266	266
R^2	0.014	0.086	0.088	0.147

Note: Standard errors clustered by subject in parentheses. Each subject contributes two observations (Prompt-AI, Data-AI). Behavioral Effects is a count variable ranging from 0 to 4. Column (4) adds demographic controls (z-scored): AI Comfort, Writing Comfort, Impatience, IQ (Ravens), Risk Inv. (Simple), Risk Inv. (Compound), Overconfidence (abs. and rel.), Big Five personality traits, Age, and Female. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

match rates: one additional behavioral pattern is associated with a three to four percentage point decrease in match rate. This suggests that for more “behavioral” subjects, their choices are less predictable from past information, regardless of its source. Moreover, their interaction term is positive and marginally significant, suggesting that this negative relationship between match rate and behavioral effects is larger in magnitude for Prompt-AI than for Data-AI.¹⁷ To put this in perspective, as seen in column (4) of Table 1, among subjects who never exhibit any behavioral effects, Data-AI and Prompt-AI have match rates of about 78% and 77%, respectively. In contrast, among subjects who exhibit all four possible behavioral effects, Prompt-AI’s match rate decreases to 61%, which is six percentage points lower than 67%, the match rate achieved by Data-AI. In other words, these results suggest that revealed-preference information is more valuable for subjects with stronger behavioral patterns, as their choices are harder to predict from stated preferences alone.

¹⁷As seen in column (4) of Table 1, these results are not driven by observable differences across subjects, as they remain similar after controlling for subjects’ demographics and survey responses.

Result 1: *AI agents perform better when given human principals’ revealed preferences rather than their stated preferences. Subjects who exhibit more behavioral patterns are harder to predict, regardless of the information source. Moreover, for these subjects, the advantage of using revealed-preference information is particularly pronounced.*

4.2 Prompt Informativeness

Why does Data-AI do better? Our analysis suggests that the gap arises because human subjects’ written prompts are not sufficiently informative. To support this view, we provide three pieces of evidence. First, we show that human-written prompts are only partially informative about subjects’ own choices, even within-sample. Second, we show that AI-generated prompts, constructed from the same choice data available to subjects, are substantially more predictive both within-sample and out-of-sample. Third, we show that the performance gap between human-written and AI-generated prompts is smaller when the two prompts are more semantically similar. Thus, these results suggest that the performance gap between AI agents is driven by human subjects’ difficulty in articulating their preferences.

First, we use subjects’ written prompts from Part I to predict their own Part I choices, which were made before the prompts were written. Intuitively, if a prompt fully captured the preference information reflected in a subject’s Part I choices, the match rate should be close to 100%. Instead, we find that the average match rate is only 70%, suggesting that these prompts are only partially informative about subjects’ preferences. Moreover, if this proxy captures prompt informativeness, then prompts that perform better in Part I should also perform better out-of-sample in Part II. Consistent with this interpretation, Prompt-AI’s match rate in Part I is positively correlated with its match rate in Part II, with a Pearson correlation coefficient of 0.43 ($p < 0.001$). As shown in Table D.2 in Appendix D, this relationship remains robust after controlling for the number of behavioral effects subjects exhibit, their demographic variables and survey responses.

Second, for each subject, we use Claude Opus 4.5 to generate a preference description based on the subject’s 13 choices in Part I, using the exact information and instructions subjects see in the second task in Part I. As a descriptive exercise, we compare the human-written and AI-generated prompts using word clouds of their most frequently occurring words (see Figure 2). We find clear differences in empha-

TABLE 2: Textual Similarity, AI Agreement, and Match Rate

	(1)	(2)	(3)
	Part II Agreement Rate	Absolute Difference in Part II Match Rate	Part II Match Rate
	AI-generated vs. Human-written prompt	Data-AI vs. Prompt-AI	Prompt-AI
SBERT Cosine Similarity	0.322*** (0.081)	-0.107** (0.042)	0.115* (0.060)
Constant	0.536 (0.054)	0.174 (0.029)	0.635 (0.039)
Observations	267	267	267
R^2	0.054	0.021	0.012

Note: HC3 robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

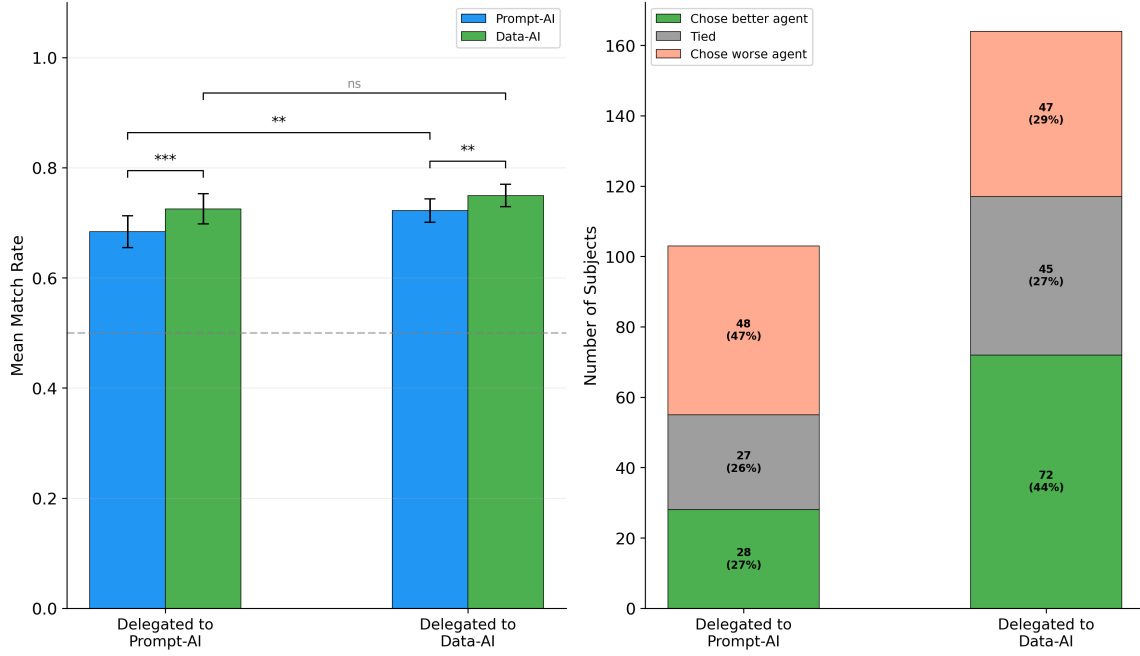
a BERT-based language model (Devlin et al., 2019). This allows us to compute the cosine similarity between the embeddings of the human-written and AI-generated prompts, with higher values indicating greater semantic similarity.¹⁸ Intuitively, if the two prompts are more semantically similar, the resulting AI agents should have more aligned predictions. Moreover, because AI-generated prompts achieve a higher match rate, which is comparable to that of Data-AI, the performance gap between Prompt-AI and Data-AI should narrow as prompt similarity increases.

Our data is consistent with these predictions. Table 2 reports OLS regressions of several measures of relative AI performance on the SBERT cosine similarity between AI-generated and human-written prompts. Each observation is at the subject level. In column (1), the dependent variable is the share of a subject’s Part II choices for which the human-written prompt and AI-generated prompt predictions agree. In column (2), the dependent variable is the absolute difference in Part II match rates between Data-AI and Prompt-AI. We find that greater textual similarity between human-written and AI-generated prompts is associated with a larger share of aligned predictions and a smaller performance gap between Data-AI and Prompt-AI. Consistent with this pattern, column (3) shows that prompt similarity is positively associated with Prompt-AI’s own predictive performance.¹⁹

¹⁸Cosine similarity measures the cosine of the angle between two vectors. It ranges from -1 to 1 , with larger values indicating that the two vectors point in more similar directions.

¹⁹Table D.3 in Appendix D shows that these relationships are robust after controlling for additional observables.

FIGURE 3: Delegation Choice and Match Rate



Note: The left panel shows the mean match rates achieved by Prompt-AI (blue) and Data-AI (green) conditional on each subject’s delegation choice. The right panel shows the number (and share) of subjects in each delegation group who chose the ex-post better agent (green), were tied (grey), or chose the ex-post worse agent (salmon). Error bars show 95% confidence intervals with standard errors clustered at the subject level. Stars denote paired t -tests comparing Prompt-AI to Data-AI within each delegation group, or t -tests comparing an AI agent’s match rate across groups. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Result 2: *The performance gap between Prompt-AI and Data-AI arises largely from humans’ difficulty in articulating preferences in text.*

4.3 Delegation

We now turn to subjects’ delegation decisions. In our data, the majority of subjects (61%) delegate to Data-AI; that is, they choose to provide the AI with their past choice data rather than their written prompts. Figure 3 assesses whether subjects delegate to the AI agent that ultimately performs better on their behalf. Panel (a) plots mean AI match rates by delegation choice. Panel (b) shows the ex-post alignment between delegation choices and realized performance at the subject-level. Conditional on a subject’s delegation choice, we refer to the chosen agent as weakly better if it achieves a higher match rate than the alternative agent, and as worse if it achieves

a strictly lower match rate. Overall, about 64% of subjects make a weakly better delegation decision.

We observe first that, on average, Data-AI achieves a higher match rate than Prompt-AI, regardless of subjects’ delegation decisions (paired t -test, $t = 3.688$, $p < 0.001$).²⁰ At the individual level, subjects who delegate to Data-AI are also about 34% more likely to choose the weakly better-performing agent. This is because among subjects who delegate to Data-AI, Data-AI is weakly better 71% of the time; in contrast, among subjects who delegate to Prompt-AI, Prompt-AI is weakly better only 53% of the time. Thus, while some subjects could potentially gain from delegating to Prompt-AI rather than Data-AI, a larger proportion—nearly half of those who delegate to Prompt-AI—would be strictly better off delegating to Data-AI instead.

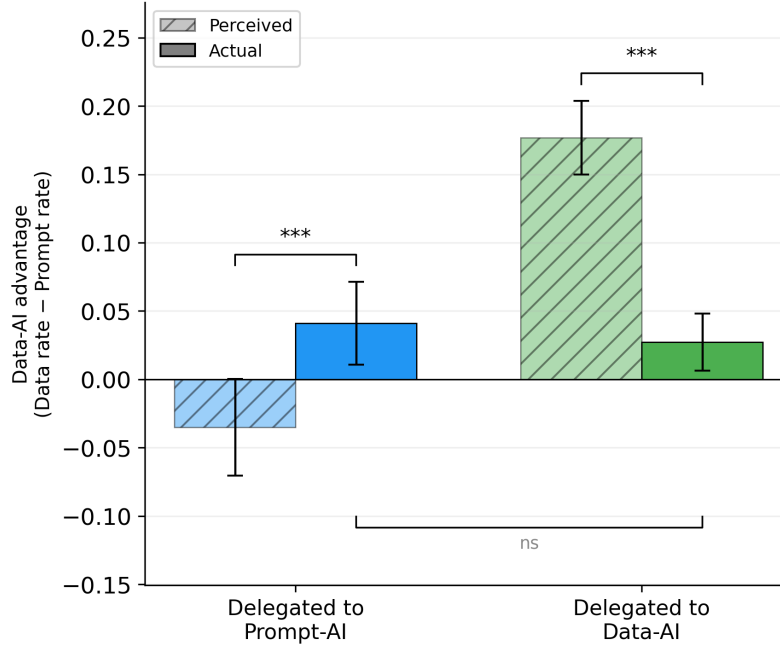
Perceived AI Performance Next, we examine how subjects’ delegation decisions relate to their perceived performance of the two AI agents. Recall that, after completing the main parts of the experiment, subjects made two incentivized guesses in Part III. They reported their guess to (i) how many of the 13 Part II lottery choices Data-AI would correctly predict, and (ii) how many Prompt-AI would correctly predict. Subjects earned an additional 25 cents for each correct guess. Overall, 82% of subjects delegate to the AI agent they expect to perform weakly better, suggesting that delegation decisions largely reflect perceived performance.

Figure 4 plots the realized and perceived differences in match rates between Data-AI and Prompt-AI, separately by delegation choice. Conditional on delegating to Prompt-AI, subjects are incorrect about the direction of relative performance: on average, they believe that Prompt-AI outperforms Data-AI, when in fact the opposite is true. Conditional on delegating to Data-AI, subjects are correct about the direction on average, but substantially overestimate the absolute difference in match rates between the two AI agents. Thus, while delegation decisions largely reflect perceived performance, these perceptions are often inaccurate in both direction and magnitude.

Finally, as suggestive evidence, we examine subjects’ free-response justifications for their delegation choices using Claude Haiku 4.5. First, we ask whether the response mentions reasons other than optimization; only 14% of subjects do so, suggesting that

²⁰Notice that subjects who delegate to Prompt-AI actually have worse prompts as indicated by their Prompt-AI performance: when comparing Prompt-AI performance conditional on delegation choice, subjects who delegate to Data-AI have a higher Prompt-AI match rate than those who delegate to Prompt-AI (t -test, $p = 0.034$).

FIGURE 4: Mean Perceived and Actual Data-AI Advantage by Delegation Choice

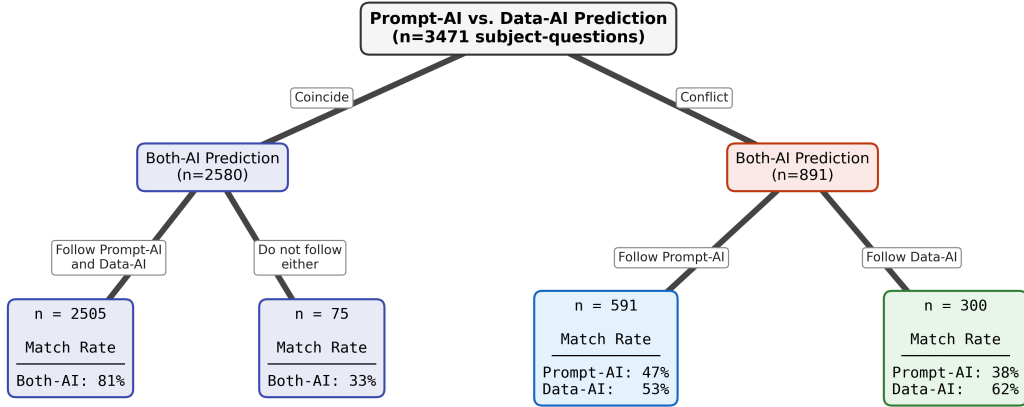


Note: Lighter bars show the mean perceived advantage (guessed Data-AI match rate minus guessed Prompt-AI match rate), while darker bars show the realized advantage. Error bars show 95% confidence intervals. Within-group brackets report paired t -tests comparing perceived to actual advantage; the lower bracket compares the actual Data-AI advantage across delegation groups. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

such concerns play a minor role in delegation decisions. Second, we check whether a response refers to either AI ability or the subject’s own ability to communicate preferences. Overall, 76% of subjects mention AI ability, while only 49% mention their own communication ability. This suggests that subjects focus more on the AI agent’s capabilities than on their own ability to communicate preferences, which may contribute to their skewed perceptions of the agents’ predictive performance.

Result 3: *More subjects prefer to provide revealed-preference rather than stated-preference information when delegating decisions to an AI agent. Although delegation decisions are largely consistent with subjects’ perceived AI performance, these perceptions are often inaccurate. Moreover, nearly half of those who provide stated-preference information would be strictly better off providing revealed-preference information instead.*

FIGURE 5: Decision Tree of Both-AI’s Choices



Note: The figure partitions subject-question pairs by whether the predictions from Prompt-AI and Data-AI coincide or conflict with each other; and, in each case, by which prediction Both-AI follows. Terminal nodes report the number of subject-question pairs and the match rates of Prompt-AI and Data-AI relative to the subject’s own choice.

4.4 Combining Revealed and Stated Preference Information

In this section we examine what happens when an AI agent is simultaneously given *both* subjects’ written prompts and their Part I choice data.²¹ We refer to this agent as **Both-AI**. We compare its performance with Prompt-AI and Data-AI to assess whether combining the two sources improves predictive accuracy.

Intuitively, giving more information to the AI agent should at least weakly improve its performance. However, we find that Both-AI outperforms Prompt-AI, but not Data-AI: its mean match rate is 72%, which is significantly lower than Data-AI’s mean match rate of 74% (paired t -test: $t = -2.50$, $p = 0.013$) but significantly higher than Prompt-AI’s mean match rate of 71% (paired t -test: $t = 2.25$, $p = 0.026$). This suggests that the additional prompt information does not always complement the choice data; rather, it sometimes leads the AI agent away from the more accurate prediction. To better understand why Both-AI fails to match Data-AI, we next examine how Both-AI behaves when Prompt-AI and Data-AI make different predictions.

Figure 5 shows which prediction Both-AI aligns with, conditional on whether

²¹This is analogous to “few-shot” prompting where the AI receives both a natural-language description of preferences and a few concrete choice examples (Sahoo et al., 2024).

Prompt-AI and Data-AI predictions coincide or conflict. Throughout our sample, their predictions coincide in 74% of subject-question pairs; in these cases, Both-AI aligns with the common prediction 97% of the time. When Prompt-AI and Data-AI predict different choices, however, Both-AI aligns with Prompt-AI 66% of the time and with Data-AI only 34% of the time, despite Data-AI being more accurate in these conflicting cases. Thus, the underperformance of Both-AI relative to Data-AI arises primarily because, when stated and revealed preference information conflict, Both-AI tends to place more weight on the less accurate stated-preference information.

To see whether such conflict resolution is AI-model specific, we replace Claude Opus 4.5 with GPT-5.4—another frontier model at the time of writing—only at the conflict-resolution stage. Specifically, we focus on the same set of 891 subject-question pairs used in Figure 5, for which Claude-based Prompt-AI and Data-AI make conflicting predictions. We then examine which prediction a GPT-based Both-AI agent follows.

We find that GPT follows Data-AI’s predictions more often than Claude does. Among the 891 conflicting cases, GPT follows Data-AI 60% of the time, compared with 34% for Claude. As a result, the mean match rate of Both-AI in these conflicting cases increases from 52% under Claude to 59% under GPT. Thus, whether an AI agent prioritizes stated or revealed preferences appears to be model-dependent. In particular, Claude’s tendency to align more closely with stated-preference information is consistent with known differences in post-training methods between these two AI models: Anthropic’s emphasis on constitutional-style alignment (Bai et al., 2022) explicitly prioritizes stated goals from a human user, whereas OpenAI uses reinforcement learning from human feedback (Ouyang et al., 2022), which does not explicitly prioritize stated preferences in the same manner.

Result 4: *Combining stated and revealed preference information does not necessarily improve predictive accuracy. When the two sources generate conflicting standalone predictions, different AI models prioritize different types of information, which may result in lower predictive accuracy.*

4.5 Robustness: GPT-5.4

As a further robustness check, we now examine whether our results depend on the choice of Anthropic’s Claude Opus 4.5 as the AI agent. This analysis allows us to

assess whether our main findings reflect broader features of how AI agents align with human principals.

Appendix E replicates our analysis by replacing Claude Opus 4.5 with GPT-5.4 as the underlying AI model. Note first that in the experiment, subjects were explicitly informed that the AI model used for payment was Claude. This may have affected the instructions they wrote, their delegation decisions, and their perceptions of AI performance. Second, this exercise differs from the analysis in Section 4.4. There, we held the Claude-based Prompt-AI and Data-AI predictions fixed and examined how a GPT-based Both-AI agent resolves conflicts between them. In this section, we instead use GPT as the underlying AI model throughout.

The results are broadly similar. For instance, Data-AI continues to outperform Prompt-AI: on average, Data-AI achieves a match rate of 74%, which is significantly higher than the 72% achieved by Prompt-AI. As before, subjects who exhibit more behavioral effects are harder to predict and benefit more from Data-AI than from Prompt-AI. Moreover, as subjects’ written prompts become more semantically similar to those generated by GPT, Prompt-AI performance also improves.

We highlight two main differences. First, the AI agent using GPT-generated prompts has a mean match rate of 71%, which no longer outperforms Prompt-AI, the AI agent using human-written prompts. This appears to be driven by the lower quality of the GPT-generated prompts rather than by the AI model itself. For example, when GPT is instead given the Claude-generated prompts, its mean match rate rises to 74%, on par with Claude’s performance using the same prompts. Second, with GPT as the AI agent, Both-AI performs approximately as well as Data-AI.²² This differs from our results with Claude, where Both-AI performs significantly worse than Data-AI. However, it is consistent with the pattern documented in Section 4.4, where replacing Claude with GPT improves performance because the latter follows Data-AI more frequently. Overall, apart from differences in conflict resolution, our results remain both qualitatively and quantitatively robust to using GPT-5.4, suggesting that they extend beyond a single AI model.

²²The mean match rate of Both-AI is 74%, which is not significantly different from Data-AI (paired t -test: $t = -0.37$, $p = 0.710$).

5 Conclusion

In this paper, we investigate whether a stated- or revealed-preference approach better enables an AI agent to implement a human principal’s preferences in the domain of choice under risk. Using an incentivized online experiment, we find that human subjects communicate their preferences more effectively through their choices than through written instructions, as subjects exhibit difficulty articulating preferences in text. At the same time, we also identify two important limitations that may affect the benefits of the revealed-preference approach. First, subjects misperceive the relative performance of AI agents, leading some to opt for the less predictive AI agent. Second, providing AI agents with more information does not necessarily improve their performance, because the outcome depends on how different AI models resolve conflicting predictions. Thus, any mechanism for instantiating economic AI agents must address both issues.

Looking forward, we note that our environment of choice under risk explicitly removes known sources of divergence between stated and revealed preference, such as present bias (O’Donoghue and Rabin, 2015) and social desirability bias (Norwood and Lusk, 2011). In these domains, stated and revealed preferences are clearly not aligned. More importantly, such a misalignment makes the evaluation of AI agents difficult as it is unclear what the human principal’s true preference is (Kleinberg et al., 2024). Furthermore, we assume that in our setting, AI agents can effectively implement the preferences they are given, which may not be true in different environments. For example, Liang (2026) proves that in higher-dimensional matching problems, infinitely many AI agent-recommended draws can be less effective than finitely many human-curated draws. Understanding the impact of stated versus revealed preferences under these types of environments and conditions would provide interesting avenues for future research.

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Appendix A Lottery Pairs

Table A.1 summarizes all lottery questions and their corresponding parameters used in the experiment. Panel A shows all 13 lotteries used in Part I of the experiment, while Panel B shows all 13 lottery pairs used in Part II. Out of each 13 lottery pairs, there are 2 Easy questions, 5 Hard Questions, and 6 Behavioral questions.

TABLE A.1: Lottery Pairs by Category: Part I and Part II

Panel A: Part I	Lottery A	Lottery B	$\mathbb{E}[B] - \mathbb{E}[A]$
<i>Easy</i>			
FOSD	(\$3.00, 0.60; \$1.00, 0.40)	(\$4.00, 0.70; \$2.00, 0.30)	+1.20
Easy	(\$1.50, 0.50; \$1.15, 0.50)	(\$1.55, 0.25; \$0.25, 0.75)	-0.75
<i>Hard</i>			
Hard 1	(\$3.85, 0.25; \$1.90, 0.75)	(\$4.70, 0.50; \$0.80, 0.50)	+0.36
Hard 2	(\$4.50, 0.50; \$0.50, 0.50)	(\$2.80, 0.25; \$2.25, 0.50; \$1.60, 0.25)	-0.28
Hard 3	(\$10.00, 0.25; \$5.25, 0.25; \$4.20, 0.25; \$0.30, 0.25)	(\$6.75, 0.25; \$5.85, 0.25; \$3.00, 0.25; \$2.70, 0.25)	-0.36
Hard 4	(\$4.05, 0.25; \$2.55, 0.25; \$1.50, 0.25; \$0.65, 0.25)	(\$4.30, 0.25; \$1.90, 0.25; \$1.60, 0.25; \$0.95, 0.25)	0.00
SOSD	(\$2.00, 1)	(\$4.00, 0.50)	0.00
<i>Behavioral</i>			
AB ($p=0.8, r=0.25$)	(\$4.00, 0.80)	(\$3.00, 1)	-0.20
AB' ($p=0.8, r=0.25$)	(\$4.00, 0.20; \$3.00, 0.75)	(\$3.00, 1)	-0.05
CD ($p=0.8, r=0.25$)	(\$4.00, 0.20)	(\$3.00, 0.25)	-0.05
AB ($p=0.5, r=0.5$)	(\$5.00, 0.50)	(\$2.00, 1)	-0.50
AB' ($p=0.5, r=0.5$)	(\$5.00, 0.25; \$2.00, 0.50)	(\$2.00, 1)	-0.25
CD ($p=0.5, r=0.5$)	(\$5.00, 0.25)	(\$2.00, 0.50)	-0.25
Panel B: Part II			
<i>Easy</i>			
FOSD	(\$2.00, 0.30; \$1.00, 0.50)	(\$3.00, 0.50; \$2.00, 0.20; \$1.00, 0.30)	+1.10
Easy	(\$15.00, 0.50; \$11.50, 0.50)	(\$15.50, 0.25; \$2.50, 0.75)	-7.50
<i>Hard</i>			
Hard 1	(\$38.50, 0.25; \$19.00, 0.75)	(\$47.00, 0.50; \$8.00, 0.50)	+3.63
Hard 2	(\$5.00, 0.50)	(\$3.00, 0.25; \$2.75, 0.50; \$1.00, 0.25)	-0.13
Hard 3	(\$9.25, 0.25; \$6.00, 0.25; \$3.50, 0.25; \$1.00, 0.25)	(\$7.30, 0.25; \$5.50, 0.25; \$3.50, 0.25; \$2.00, 0.25)	-0.36
Hard 4	(\$3.75, 0.25; \$3.00, 0.25; \$2.00, 0.25)	(\$4.50, 0.25; \$2.50, 0.25; \$1.25, 0.25; \$0.50, 0.25)	0.00
SOSD	(\$3.00, 0.25; \$2.00, 0.50; \$1.00, 0.25)	(\$3.00, 0.50; \$1.00, 0.50)	0.00
<i>Behavioral</i>			
AB ($p=10/11, r=0.11$)	(\$3.00, 0.95)	(\$4.00, 0.91)	+0.79
AB' ($p=10/11, r=0.11$)	(\$3.00, 0.95)	(\$4.00, 0.10; \$3.00, 0.89)	+0.22
CD ($p=10/11, r=0.11$)	(\$3.00, 0.11)	(\$4.00, 0.10)	+0.07
AB ($p=0.3, r=0.5$)	(\$1.00, 0.95)	(\$5.00, 0.30)	+0.55
AB' ($p=0.3, r=0.5$)	(\$1.00, 0.95)	(\$5.00, 0.15; \$1.00, 0.50)	+0.30
CD ($p=0.3, r=0.5$)	(\$1.00, 0.50)	(\$5.00, 0.15)	+0.25

Note: lotteries are represented as $(x_1, p_1; \dots, x_n, p_n)$ where each outcome x_i occurs with probability $p_i \geq 0$ and $\sum_{i=1}^n p_i = 1$. To simplify notation, we omit the null outcome of 0 and use (x, p) to denote the prospect $(x, p; 0, 1 - p)$.

Appendix B AI System Prompts

This section reproduces the system prompts used for the Prompt-AI and Data-AI agents verbatim. Both agents additionally receive the 13 post-prompt lottery choices as a user message (see Section 3).

Prompt-AI System Prompt

You are helping a participant in an economics experiment make decisions between lottery pairs. The participant has provided you with instructions about their preferences.

Based on the participant's instructions below, you will make choices between pairs of lotteries on their behalf. For each choice, respond with ONLY "A" or "B" to indicate your selection.

PARTICIPANT'S INSTRUCTIONS:

"""

[PARTICIPANT'S WRITTEN PROMPT -- inserted verbatim]

"""

You will now be presented with lottery choices. After reasoning through each choice, you MUST end your response with a JSON object containing your choices in this exact format:

```
{"post_1": "X", "post_2": "X", "post_3": "X", "post_4": "X",  
  "post_5": "X", "post_6": "X", "post_7": "X", "post_8": "X",  
  "post_9": "X", "post_10": "X", "post_11": "X", "post_12": "X",  
  "post_13": "X"}
```

Replace X with your actual choices. The JSON must be the last thing in your response.

Data-AI System Prompt

You are an AI assistant helping a participant in an economics experiment make decisions between lottery pairs.

Based on the participant's previous choices shown below, infer their preferences and make similar choices for new lottery pairs.

You will now be presented with lottery choices. After reasoning through each choice, you MUST end your response with a JSON object containing your choices in this exact format:

```
{"post_1": "A", "post_2": "B", ..., "post_13": "A"}
```

Replace A/B with your actual choices. The JSON must be the last thing in your response.

PARTICIPANT'S PREVIOUS CHOICES (Questions 1-13):

- Q1: Chose Lottery [X] (A: [description], B: [description])
- Q2: Chose Lottery [X] (A: [description], B: [description])
- ...
- [All 13 pre-prompt choices with full lottery descriptions]

Appendix C Structural Exercise

In this section, we describe our structural exercise. For each subject, we estimate a constant relative risk aversion (CRRA) expected utility model with a logit choice error using the 13 lottery choices in Part I. Given a lottery $L = (x_i, p_i)_{i=1}^n$, the expected utility is

$$EU(L; \rho) = \sum_{i=1}^n p_i u(x_i; \rho), \text{ where } u(x; \rho) = \begin{cases} \frac{(x+w)^{1-\rho}}{1-\rho} & \text{if } \rho \neq 1, \\ \ln(x+w) & \text{if } \rho = 1, \end{cases}$$

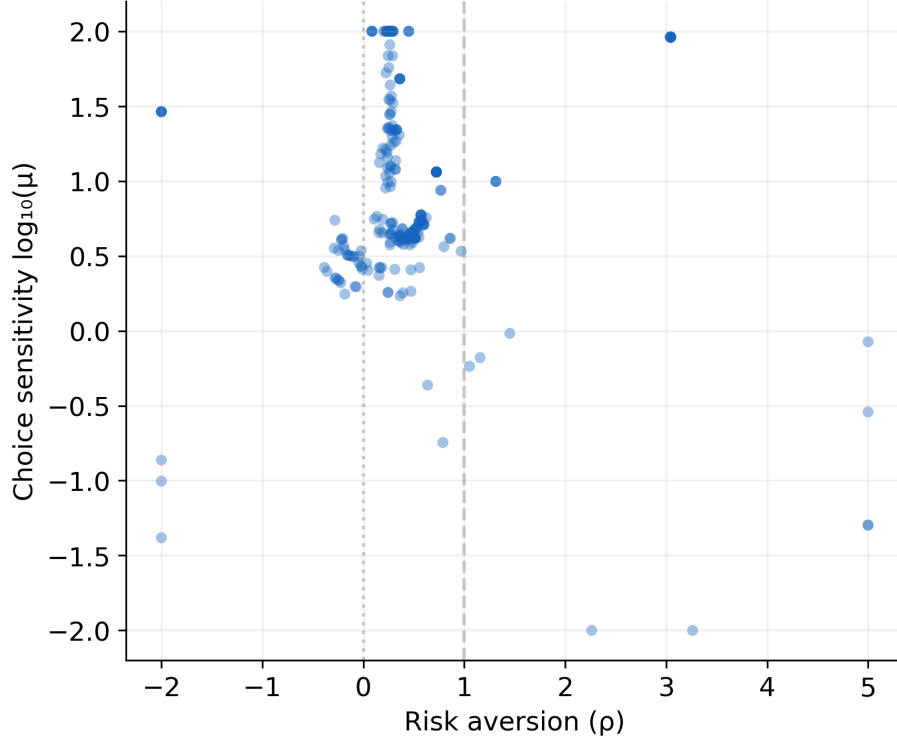
and ρ is the coefficient of relative risk aversion, $w = 0.01$ is a small wealth offset that ensures the utility function is defined at zero payoffs. For a lottery pair (A, B) , the probability of lottery A being chosen by the agent with CRRA parameter ρ is thus

$$\mathbb{P}(A; \rho, \mu) = \frac{1}{1 + e^{-\mu[EU(A; \rho) - EU(B; \rho)]}},$$

where $\mu \geq 0$ is a precision parameter. As $\mu \rightarrow \infty$, the agent deterministically chooses the higher-EU lottery. As $\mu \rightarrow 0$, their choices approach uniform random choice.

For each subject, we observe 13 choices in Part I and use maximum likelihood to

FIGURE C.1: Scatter plot of estimated $\hat{\mu}$ and $\hat{\rho}$ values per subject, based on their Part I choices.



Note: Dashed lines at $\rho = 0$ and $\rho = 1$ indicate risk neutrality and log utility, respectively.

estimate their individual (ρ, μ) . The log-likelihood is

$$\ell(\rho, \mu) = \sum_{k=1}^{13} [\mathbb{1}_{c_k=A_k} \ln \mathbb{P}(A_k; \rho, \mu) + \mathbb{1}_{c_k=B_k} \ln \mathbb{P}(B_k; \rho, \mu)],$$

where $c_k \in \{A_k, B_k\}$ is the observed choice from binary lottery menu $k = 1, \dots, 13$. We then optimize over the bounded parameter space $\rho \in [-2, 5]$, $\mu \in [0.01, 100]$ using L-BFGS-B, with a grid of 24 starting points ($\rho_0 \in \{-0.5, 0, 0.5, 1, 2, 3\}$, $\mu_0 \in \{0.1, 1, 5, 20\}$) to mitigate local optima. Finally, we retain the solution with the largest log-likelihood.

Figure C.1 shows a scatter plot of our estimated parameters for each subject. Most subjects are moderately risk averse: 70% have estimated CRRA parameters ρ between zero and one. Among these subjects, the median ρ is 0.34, and the median precision parameter is 5.9.

We then use these estimated risk parameters to predict subjects’ choices in Part II; we refer to these as predictions from the expected utility (EU) model. As described in the main text, the EU model achieves an average match rate of about 75%, similar to that of Data-AI. Moreover, at the subject-question level, EU predictions are more closely aligned with Data-AI than with Prompt-AI, matching the former 80% of the time and the latter 73% of the time. Across question categories, we also find that the EU model is less predictive for hard and behavioral questions: the average match rates for easy, behavioral, and hard are 99%, 72%, and 70%, respectively.

Appendix D Additional Analysis

In this section, we first describe the additional set of controls that we elicited in Part III of the experiment. We then present three additional regression tables. Table D.1 reports the full specification corresponding to column (4) of Table 1. Table D.2 extends the discussion in Section 4.2 by regressing Prompt-AI’s Part II match rate on its Part I match rate while adding these controls. Finally, Table D.3 extends Table 2 by adding the same set of controls.

- *AI Comfort* and *Writing Comfort* are single 1–7 Likert items elicited at the end of the experiment: “How comfortable are you with using AI tools (e.g., ChatGPT, Claude, Copilot)?” and “How comfortable are you with writing instructions or explanations for others?”, respectively, with endpoints labeled “Not at all comfortable” and “Very comfortable”.
- *Impatience* is a single 1–7 Likert item: “How would you rate your level of impatience in general?”, with endpoints labeled “Very patient” and “Very impatient”.
- *IQ (Raven’s)* is the number of correct answers on a 6-item matrix reasoning test from the International Cognitive Ability Resource (ICAR; [Condon and Revelle, 2014](#)) (score 0–6). Subjects receive 20¢ for each correct answer.
- *Measures of Overconfidence* consist of two types. The first is the difference between the subject’s self-reported guess of their own score and their actual IQ test score, which captures absolute overconfidence. The second is the difference

between the subject’s self-reported guess of their own score and their performance relative to other Prolific subjects, which captures relative overconfidence. Subjects receive 25¢ for each correct answer.

- *Risk Inv. (Simple)* and *Risk Inv. (Compound)* are the amounts (in tokens, from 0 to 100) invested in two variants of the investment task in [Gneezy and Potters \(1997\)](#). Subjects are endowed with 100 tokens, and each token is worth 0.5¢. Subjects can choose to invest their tokens into a safe asset, which leaves the invested amount unchanged, or in a risky asset, which multiplies the invested amount by 2.5 if the investment succeeds. The simple task resolves uncertainty in a single coin flip whereas the compound task involves the same reduced distribution over payoffs, implemented by a compounded lottery: in stage one, there is a 25% chance of success, a 25% chance of failure, and a 50% chance of proceeding to the second stage. In the second stage, uncertainty is again resolved by a coin flip. Subjects are paid for each task based on their remaining balance of tokens after the uncertainty is resolved.
- The “Big Five” personality traits (*Extraversion*, *Agreeableness*, *Conscientiousness*, *Emotional Stability*, *Openness*) are measured using the Ten-Item Personality Inventory (TIPI; [Gosling et al., 2003](#)), with each trait based on two 1–7 Likert items. There are two questions for each trait, one of which is reverse-coded. For example, *Extraversion* is measured as the average of the response to “I see myself as extraverted, enthusiastic” and the reverse-coded response to “I see myself as reserved, quiet”.

TABLE D.1: OLS: Match Rate (Prompt-AI and Data-AI)

	Match rate
Data-AI (vs Prompt-AI)	0.013 (0.014)
Behavioral Effects	-0.040*** (0.008)
Data-AI $\times$ Behavioral Effects	0.012* (0.007)
AI Comfort	-0.009 (0.008)
Writing Comfort	0.004 (0.009)
Impatience	0.020** (0.008)
IQ (Ravens)	0.020** (0.009)
Risk Inv. (Simple)	0.007 (0.011)
Risk Inv. (Compound)	-0.006 (0.011)
Overconfidence (abs.)	-0.008 (0.009)
Overconfidence (rel.)	0.011 (0.009)
Extraversion	-0.013* (0.007)
Age	0.011 (0.008)
Female	-0.008 (0.008)
Constant	0.770 (0.014)
Observations	532
Subjects	266
R^2	0.147

Note: Standard errors clustered by subject are reported in parentheses. Behavioral effects are included as a count variable ranging from 0 to 4, indicating the number of behavioral patterns exhibited by a subject. All regressors, except behavioral effects, the indicator for Data-AI, and their interaction, are standardized as z-scores. Agreeableness, Conscientiousness, Emotional Stability, and Openness are included as controls (none are significant) but not reported. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE D.2: OLS: Prompt-AI Match Rate Across Parts

	Prompt-AI Match Rate (Part II)
Prompt-AI Match Rate (Part I)	0.052*** (0.009)
Behavioral Effects	-0.023*** (0.008)
AI Comfort	0.005 (0.010)
Writing Comfort	-0.002 (0.011)
Impatience	0.006 (0.009)
IQ (Ravens)	0.021* (0.011)
Risk Inv. (Simple)	-0.004 (0.015)
Risk Inv. (Compound)	0.014 (0.014)
Overconfidence (abs.)	0.001 (0.012)
Overconfidence (rel.)	0.011 (0.012)
Extraversion	-0.004 (0.010)
Age	0.019** (0.008)
Female	-0.011 (0.009)
Constant	0.744 (0.015)
Observations	266
R^2	0.261

Notes: HC3 robust standard errors in parentheses. Behavioral effects are included as a count variable ranging from 0 to 4, indicating the number of behavioral patterns exhibited by a subject. All regressors, except behavioral effects and Prompt-AI match rates, are standardized as z-scores. Agreeableness, Conscientiousness, Emotional Stability, and Openness are included as controls (none are significant) but not reported. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE D.3: Textual Similarity, AI Agreement, and Match Rate

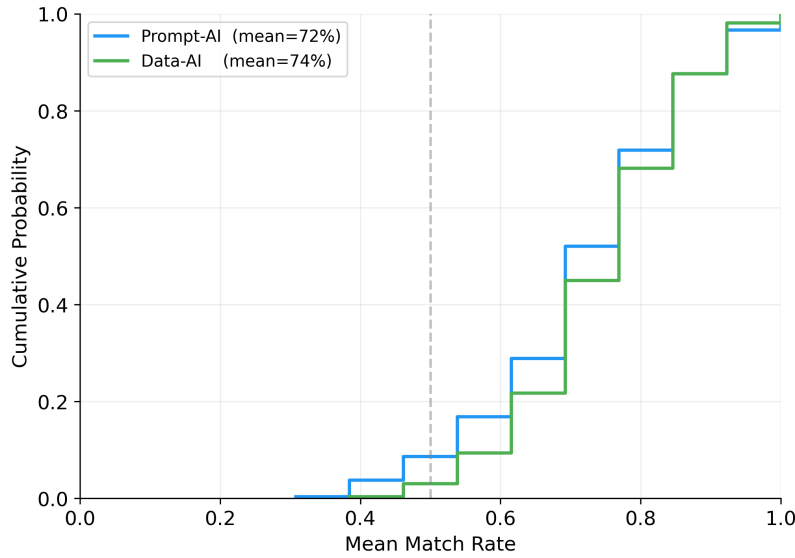
	(1)	(2)	(3)
	Part II Agreement Rate	Absolute Difference in Part II Match Rate	Part II Match Rate
	AI-generated vs. Human-written prompt	Data-AI vs. Prompt-AI	Prompt-AI
SBERT Cosine Similarity	0.322*** (0.087)	-0.124*** (0.048)	0.123* (0.066)
Behavioral Effects	-0.032*** (0.010)	0.008 (0.005)	-0.039*** (0.008)
AI Comfort	-0.010 (0.013)	-0.011 (0.007)	0.000 (0.010)
Writing Comfort	-0.008 (0.014)	0.012 (0.008)	-0.002 (0.011)
Impatience	0.011 (0.015)	-0.002 (0.009)	0.011 (0.011)
IQ (Ravens)	-0.019 (0.016)	-0.005 (0.009)	0.017 (0.012)
Risk Inv. (Simple)	-0.027 (0.019)	0.016 (0.013)	0.000 (0.016)
Risk Inv. (Compound)	0.003 (0.018)	-0.011 (0.013)	0.011 (0.015)
Overconfidence (abs.)	-0.010 (0.018)	-0.000 (0.011)	-0.006 (0.012)
Overconfidence (rel.)	-0.025 (0.018)	-0.003 (0.011)	0.010 (0.012)
Extraversion	0.009 (0.014)	0.000 (0.009)	-0.010 (0.010)
Age	-0.019 (0.013)	-0.011 (0.007)	0.019** (0.010)
Female	-0.012 (0.012)	0.005 (0.008)	-0.015 (0.010)
Constant	0.587 (0.062)	0.173 (0.035)	0.690 (0.045)
Observations	266	266	266
R^2	0.186	0.100	0.166

Notes: HC3 robust standard errors in parentheses. Behavioral effects are included as a count variable ranging from 0 to 4, indicating the number of behavioral patterns exhibited by a subject. All regressors, except SBERT cosine similarity and behavioral effects, are standardized as z-scores. Agreeableness, Conscientiousness, Emotional Stability, and Openness are included as controls but not reported. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix E GPT-5.4 Analysis

This section replicates the main analysis using GPT-5.4 (gpt-5.4) as the underlying AI model instead of Claude Opus 4.5. Figure E.1 and Table E.1 replicate the analysis in Section 4.1. For the analysis in Section 4.2, Table E.2 and Table E.3 replicate Table D.2 and Table 2, respectively. Figure E.2 and Figure E.3 replicate the figures in Section 4.3. Finally, Figure E.4 replicates Figure 5.

FIGURE E.1: Comparison of Prompt-AI and Data-AI match rates (GPT-5.4).



Note: The empirical CDF of per-subject match rates are shown for Prompt-AI (blue) and Data-AI (green). The dashed line at 50% is the expected match rate of random guessing. Error bars show 95% confidence intervals with standard errors clustered at the subject level. Stars denote paired t -tests comparing Data-AI to Prompt-AI within each category. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

TABLE E.1: OLS: Match Rate (GPT-5.4)

	Match rate			
	(1)	(2)	(3)	(4)
Data-AI (vs Prompt-AI)	0.026*** (0.009)	0.026*** (0.009)	-0.021 (0.013)	-0.021 (0.014)
Behavioral Effects		-0.022*** (0.005)	-0.037*** (0.007)	-0.042*** (0.007)
Data-AI $\times$ Behavioral Effects			0.030*** (0.008)	0.030*** (0.008)
Constant	0.717 (0.009)	0.752 (0.013)	0.776 (0.014)	0.784 (0.014)
Controls	No	No	No	Yes
Observations	532	532	532	532
Subjects	266	266	266	266
R^2	0.009	0.046	0.063	0.132

Note: Standard errors clustered by subject in parentheses. Each subject contributes two observations (Prompt-AI, Data-AI). Behavioral Effects is a count variable ranging from 0 to 4. Column (4) adds demographic controls (z-scored): AI Comfort, Writing Comfort, Impatience, IQ (Ravens), Risk Inv. (Simple), Risk Inv. (Compound), Overconfidence (abs. and rel.), Big Five personality traits, Age, and Female. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE E.2: OLS: Prompt-AI Match Rate Across Parts (GPT-5.4)

	Prompt-AI Match Rate (Part II)
Prompt-AI Match Rate (Part I)	0.046*** (0.009)
Behavioral Effects	-0.028*** (0.008)
AI Comfort	-0.002 (0.010)
Writing Comfort	-0.000 (0.012)
Impatience	0.002 (0.010)
IQ (Ravens)	0.010 (0.012)
Risk Inv. (Simple)	0.007 (0.016)
Risk Inv. (Compound)	0.009 (0.016)
Overconfidence (abs.)	-0.008 (0.012)
Overconfidence (rel.)	0.010 (0.013)
Extraversion	-0.005 (0.010)
Age	0.008 (0.009)
Female	-0.014 (0.010)
Constant	0.762 (0.015)
Observations	266
R^2	0.238

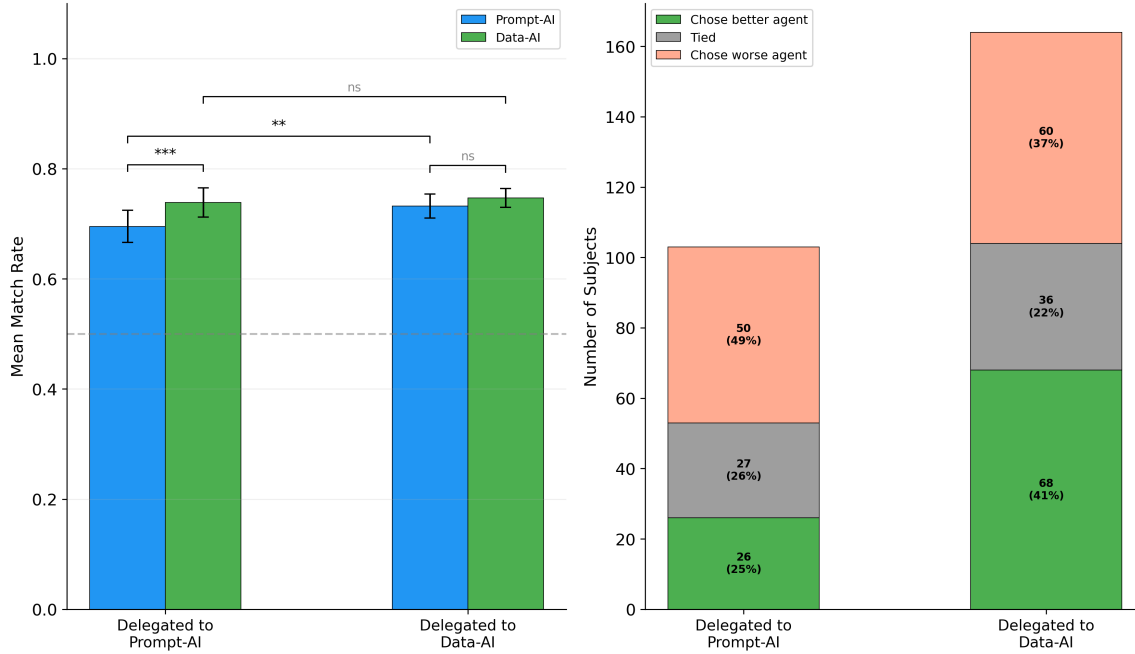
Note: HC3 robust standard errors in parentheses. Behavioral Effects is included as a count variable ranging from 0 to 4, indicating the number of behavioral patterns exhibited by a subject. All regressors except Behavioral Effects are standardized as z-scores. Agreeableness, Conscientiousness, Emotional Stability, and Openness are included as controls but not reported. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE E.3: Textual Similarity, AI Agreement, and Match Rate (GPT-5.4)

	(1)	(2)	(3)
	Part II Agreement Rate	Absolute Difference in Part II Match Rate	Part II Match Rate
	AI-generated vs. Human-written prompt	Data-AI vs. Prompt-AI	Prompt-AI
SBERT Cosine Similarity	0.200*** (0.074)	-0.089** (0.044)	0.125** (0.062)
Constant	0.619 (0.048)	0.166 (0.028)	0.641 (0.039)
Observations	267	267	267
R^2	0.026	0.016	0.013

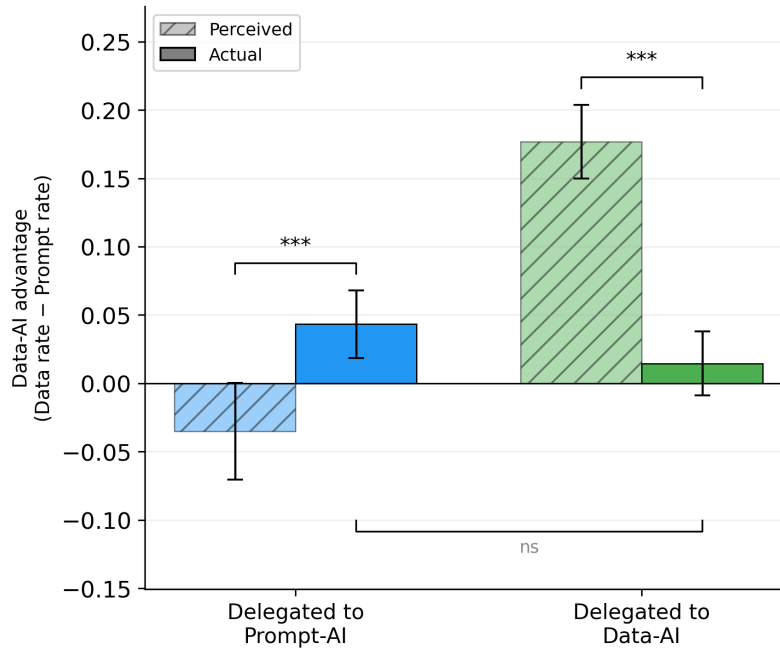
Note: HC3 robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

FIGURE E.2: Delegation Choice and Match Rate (GPT-5.4)



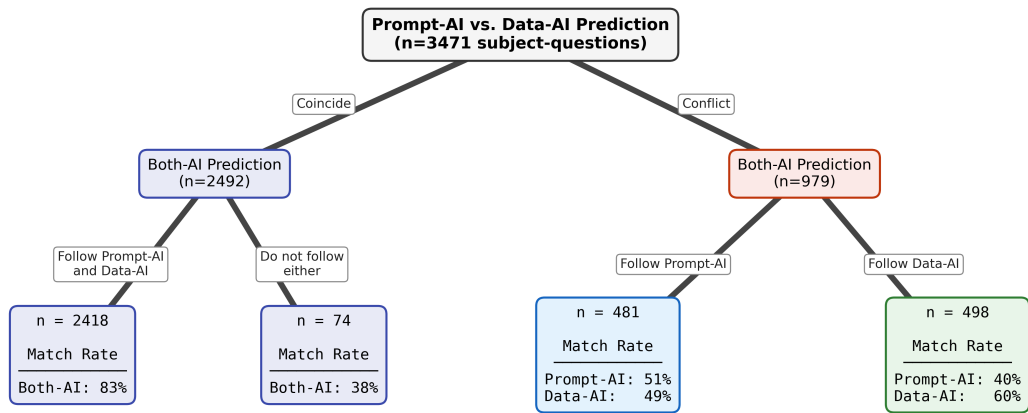
Note: The left panel shows the mean match rates achieved by Prompt-AI (blue) and Data-AI (green) conditional on each subject's delegation choice. The right panel shows the number (and share) of subjects in each delegation group who chose the ex-post better agent (green), were tied (grey), or chose the ex-post worse agent (salmon). Error bars show 95% confidence intervals with standard errors clustered at the subject level. Stars denote paired t -tests comparing Prompt-AI to Data-AI within each delegation group, or t -tests comparing an AI agent's match rate across groups. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

FIGURE E.3: Mean Perceived and Actual Data-AI Advantage by Delegation Choice (GPT-5.4)



Note: Lighter bars show the mean perceived advantage (guessed Data-AI match rate minus guessed Prompt-AI match rate), while darker bars show the realized advantage. Error bars show 95% confidence intervals. Within-group brackets report paired t -tests comparing perceived to actual advantage; the lower bracket compares the actual Data-AI advantage across delegation groups. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

FIGURE E.4: Decision Tree of Both-AI's Choices (GPT-5.4)



Note: The figure partitions subject-question pairs by whether the predictions from Prompt-AI and Data-AI coincide or conflict with each other; and, in each case, by which prediction Both-AI follows. Terminal nodes report the number of subject-question pairs and the match rates of Prompt-AI and Data-AI relative to the subject's own choice.